

Class 29: The Characteristic Polynomial

Definition: If A is a square matrix, we define an characteristic polynomial for A to be $\det(A - \lambda I)$

The characteristic equation is $\det(A - \lambda I) = 0$.

The eigenvalues for A are the zeros for the characteristic equation.

Practice: Find the characteristic polynomial for $A = \begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix}$

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 5 \\ 2 & 4-\lambda \end{vmatrix} = (1-\lambda)(4-\lambda) - 10 = \lambda^2 - 5\lambda - 6$$

Set this equal to zero and solve for the two eigenvalues.

$$\lambda^2 - 5\lambda - 6 = 0 \quad \text{so} \quad (\lambda - 6)(\lambda + 1) = 0 \quad \text{so} \quad \lambda = 6 \text{ or } -1$$

Find an eigenvector for $\lambda = 6$ by finding a basis for the null space of the matrix $A - 6I$

$$A - 6I = \begin{bmatrix} -5 & 5 \\ 2 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \quad \begin{matrix} x_1 = x_2 \\ x_2 = x_2 \end{matrix} \quad \text{Basis vector} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Find an eigenvector for $\lambda = -1$ by finding a basis for the null space of the matrix $A + I$

$$A + I = \begin{bmatrix} 2 & 5 \\ 2 & 5 \end{bmatrix} \sim \begin{bmatrix} 2 & 5 \\ 0 & 0 \end{bmatrix} \quad \begin{matrix} 2x_1 = -5x_2 \\ x_2 = x_2 \end{matrix} \quad \text{Basis vector} = \begin{bmatrix} -5/2 \\ 1 \end{bmatrix} \\ \text{or} \quad \begin{bmatrix} -5 \\ 2 \end{bmatrix}$$

These two vectors are linearly independent so they form a basis for $\mathbb{R}^2$.

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 2 \end{bmatrix} \right\}$$

The vector $\begin{bmatrix} -3 \\ 11 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} -5 \\ 2 \end{bmatrix}$

Calculate $A^5 \begin{bmatrix} -3 \\ 11 \end{bmatrix} = A^5 \left(7 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} -5 \\ 2 \end{bmatrix} \right) = 7 \cdot A^5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2 A^5 \begin{bmatrix} -5 \\ 2 \end{bmatrix}$

$$= 7 \cdot 6^5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2 (-1)^5 \begin{bmatrix} -5 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \cdot 6^5 - 2(-5) \\ 7 \cdot 6^5 - 2(2) \end{bmatrix} = \begin{bmatrix} 54442 \\ 54428 \end{bmatrix}$$

Let $P = \begin{bmatrix} 1 & -5 \\ 1 & 2 \end{bmatrix}$. Then $P^{-1}AP = \begin{bmatrix} 6 & 0 \\ 0 & -1 \end{bmatrix}$

Definition: Matrix A is similar to matrix B if there is some invertible matrix P such that $P^{-1}AP = B$

So in the example $A = \begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix}$ is similar to $\begin{bmatrix} 6 & 0 \\ 0 & -1 \end{bmatrix}$

If A is similar to B , then it will be true that B is also similar to A . We say that A and B are similar to each other.

Notice in the example that A and B had the same eigenvalues. That is always true for similar matrices.

Theorem: If A and B are similar, then A and B have the same characteristic polynomial and hence the same eigenvalues.

Proof:

Let A and B be similar matrices. This means $B = P^{-1}AP$ for some matrix P . Now the characteristic polynomial for B is $\det(B - \lambda I)$.

$$\text{But } B - \lambda I = P^{-1}AP - \lambda P^{-1}P = P^{-1}AP - P^{-1}\lambda IP = P^{-1}(A - \lambda I)P.$$

$$\text{Therefore } \det(B - \lambda I) = \det(P^{-1}(A - \lambda I)P) = \det P^{-1} \cdot \det(A - \lambda I) \cdot \det P$$
$$= \frac{1}{\det P} \cdot \det P \cdot \det(A - \lambda I) = \det(A - \lambda I)$$

So char poly for B = char poly for A .

Big theorem: If $\mathbf{v}_1, \dots, \mathbf{v}_r$ are eigenvectors that correspond to distinct eigenvalues for matrix A , then the set $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent.

Proof by contradiction:

Suppose for contradiction that $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly dependent. Then at least one vector, $\mathbf{v}_p$, is a combination of the preceding vectors in the list.

Let p be the smallest subscript for which this is true. This would mean $\{\mathbf{v}_1, \dots, \mathbf{v}_{p-1}\}$ is linearly independent and $\mathbf{v}_p = c_1 \mathbf{v}_1 + \dots + c_{p-1} \mathbf{v}_{p-1}$

Multiply both sides by A .

$$A \vec{v}_p = A [c_1 \mathbf{v}_1 + \dots + c_{p-1} \mathbf{v}_{p-1}]$$

$$\lambda_p \vec{v}_p = c_1 \lambda_1 \mathbf{v}_1 + \dots + c_{p-1} \lambda_{p-1} \mathbf{v}_{p-1}$$

$$\text{But } \lambda_p \vec{v}_p = \lambda_p (c_1 \mathbf{v}_1 + \dots + c_{p-1} \mathbf{v}_{p-1}) = c_1 \lambda_p \mathbf{v}_1 + \dots + c_{p-1} \lambda_p \mathbf{v}_{p-1}$$

$$\mathbf{0} = \lambda_p \vec{v}_p - \lambda_p \mathbf{v}_p = \underline{c_1 (\lambda_1 - \lambda_p) \mathbf{v}_1} + \dots + \underline{c_{p-1} (\lambda_{p-1} - \lambda_p) \mathbf{v}_{p-1}}$$

Since $\{\mathbf{v}_1, \dots, \mathbf{v}_{p-1}\}$ is Lin ind we know the only set of weights that produces $\vec{0}$ is if $c_i (\lambda_i - \lambda_p) = 0$ for each i . But the λ 's are distinct so $\lambda_i - \lambda_p \neq 0$
Forces $c_i = 0$ for each c_i

$\therefore \mathbf{v}_p = 0 \mathbf{v}_1 + \dots + 0 \mathbf{v}_{p-1} = \vec{0}$. This says $\mathbf{v}_p$ is not an e-vector

Thus we get a contradiction. So $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ cannot be linearly dependent. It must be a linearly independent set!

